

NAVISAFE

A Safety-First Smart Navigation System.

Guide

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INTRODUCTION

- Road accidents are a leading cause of injuries and fatalities worldwide.
- Conventional navigation apps focus mainly on shortest time or distance.
- Such systems ignore road safety factors like accident-prone areas (black spots).
- NaviSafe aims to provide safer route recommendations using data analytics and AI.

PROBLEM STATEMENT

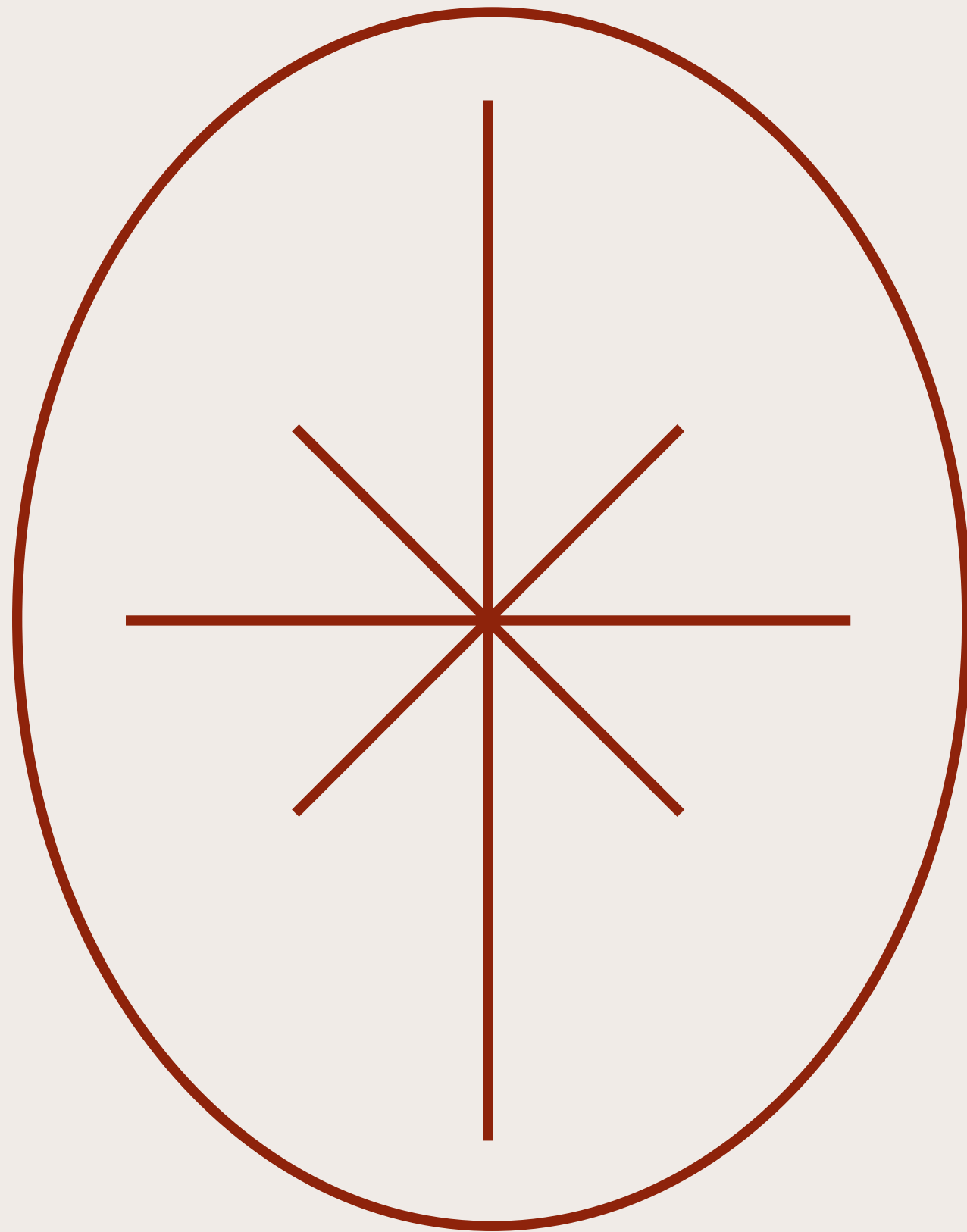
To transform conventional navigation into a safety-first system by prioritizing accident risk over shortest distance.

LITERATURE REVIEW

Name Of Paper	Year	Methodology	Result	Limitation
Accident Prediction Using Machine Learning Techniques	2019	Historical road accident data analyzed using Machine Learning algorithms such as Random Forest and SVM to predict accident-prone locations. Feature selection was performed based on road, weather, and traffic conditions.	Achieved improved accuracy in identifying high-risk accident zones compared to traditional statistical methods	Does not support real-time navigation or route recommendation

Name Of Paper	Year	Methodology	Result	Limitation
GIS-Based Identification of Road Accident Black Spots	2020	Geographic Information System (GIS) techniques were used to spatially analyze accident frequency and severity to identify black spots on road networks.	Effectively identified accident-prone road segments with high spatial accuracy	Lacks integration with navigation systems and real-time user guidance.
Risk-Aware Route Planning for Intelligent Transportation Systems	2021	Developed a risk-weighted routing algorithm that considers accident probability along with distance and travel time..	Safer routes reduced exposure to high-risk road segments in simulations..	Tested only in simulated environments, not real-world deployment..

Name Of Paper	Year	Methodology	Result	Limitation
AI-Based Navigation Systems for Road Safety Enhancement	2022	Combined data analytics and machine learning with map APIs to generate safety-aware navigation paths.	Demonstrated improvement in road safety awareness and route selection.	Limited scalability and dependency on external APIs.
Handling OpenStreetMap Georeferenced Data for Route Planning	2022	Collected and processed OpenStreetMap (OSM) geospatial data. Imported data into a PostGIS spatial database. Developed an enhanced road network model. Applied A-Star algorithm for multi-criteria route planning.	Successfully demonstrated an efficient framework for processing OSM data for route planning. Enabled multi-criteria routing based on enhanced geospatial attributes.	Highly dependent on the accuracy and completeness of OpenStreetMap data. Limited real-world validation and user-level testing. Multi-criteria weighting may vary based on user preferences and context.



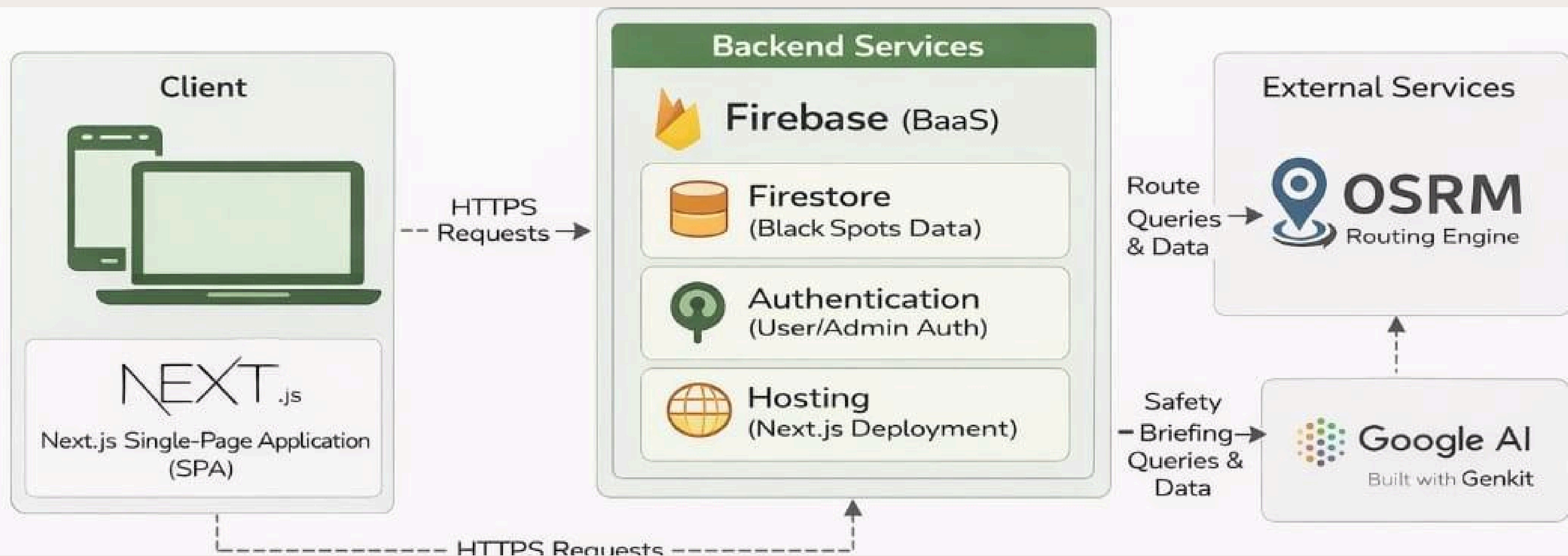
GAP IDENTIFIED

- Most studies focus on accident analysis, not real-time navigation.
- Limited integration of black spot data with popular map services.
- Lack of user-focused safety-based route selection.
- Need for a practical, low-cost, AI-driven safer navigation system.

PROPOSED SOLUTION

- Analyze historical accident datasets.
- Identify black spots based on frequency and severity.
- Assign risk scores to road segments.
- Integrate risk data with map services.
- Recommend routes with minimal risk exposure.

SYSTEM ARCHITECTURE



ARCHITECTURE OVERVIEW

NaviSafe follows a Web-Based Client-Server Architecture using Firebase as backend services. The system integrates accident data analytics with map services to provide safer route recommendation

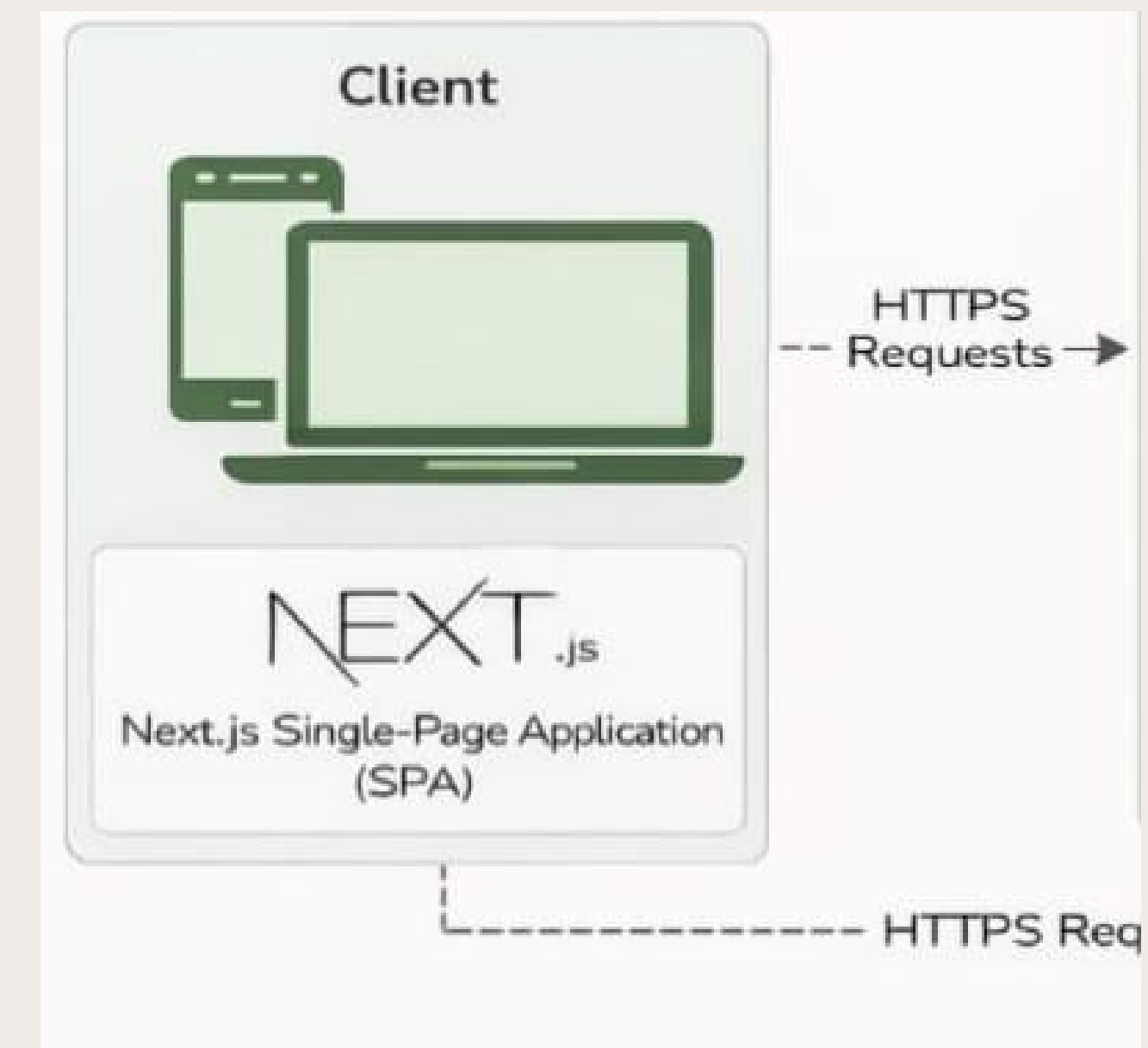
ARCHITECTURE COMPONENTS

User Layer

- Accesses the system through a web browser
- Enters source and destination
- Views safest route with risk indicators

Frontend Layer

- Developed using Next.js (TypeScript)
- Styled with Tailwind CSS Handles user interaction
- Sends route requests to backend
- Displays route and black spot alerts



Backend Layer (Firebase)

- Uses Firebase Firestore (NoSQL)
- Stores accident and risk data
- Handles real-time data retrieval
- Applies Firebase security rules

Risk Analysis Module

- Processes historical accident data
- Identifies accident-prone areas (Black Spots)
- Assigns risk scores to road segments



Route Recommendation Engine

- Combines shortest path with risk score
- Selects minimum-risk route
- Sends safest route to frontend

Map Integration

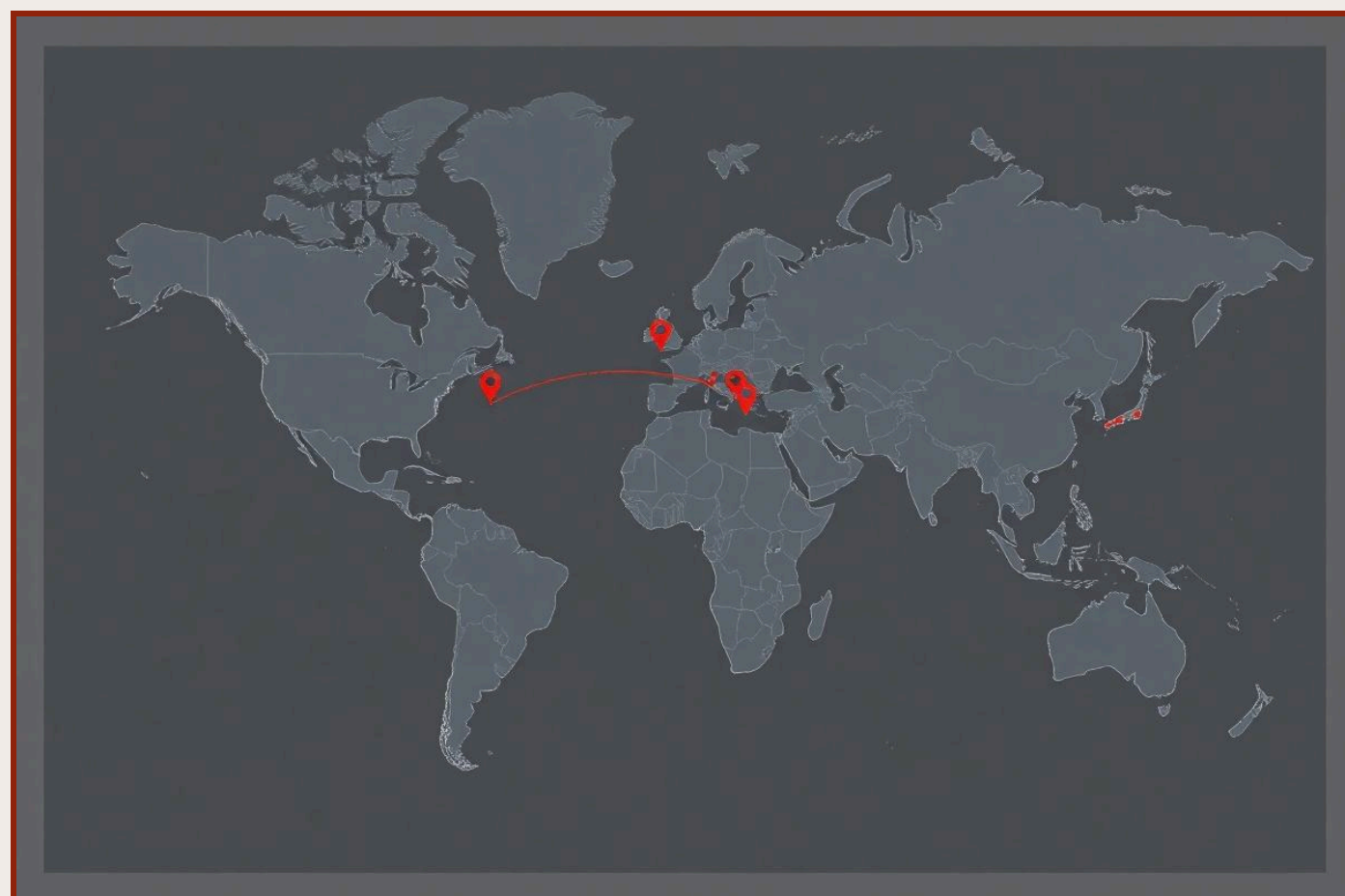
- Integrates with map services
- Displays safest route visually
- Highlights high-risk areas



MODULE DESCRIPTION

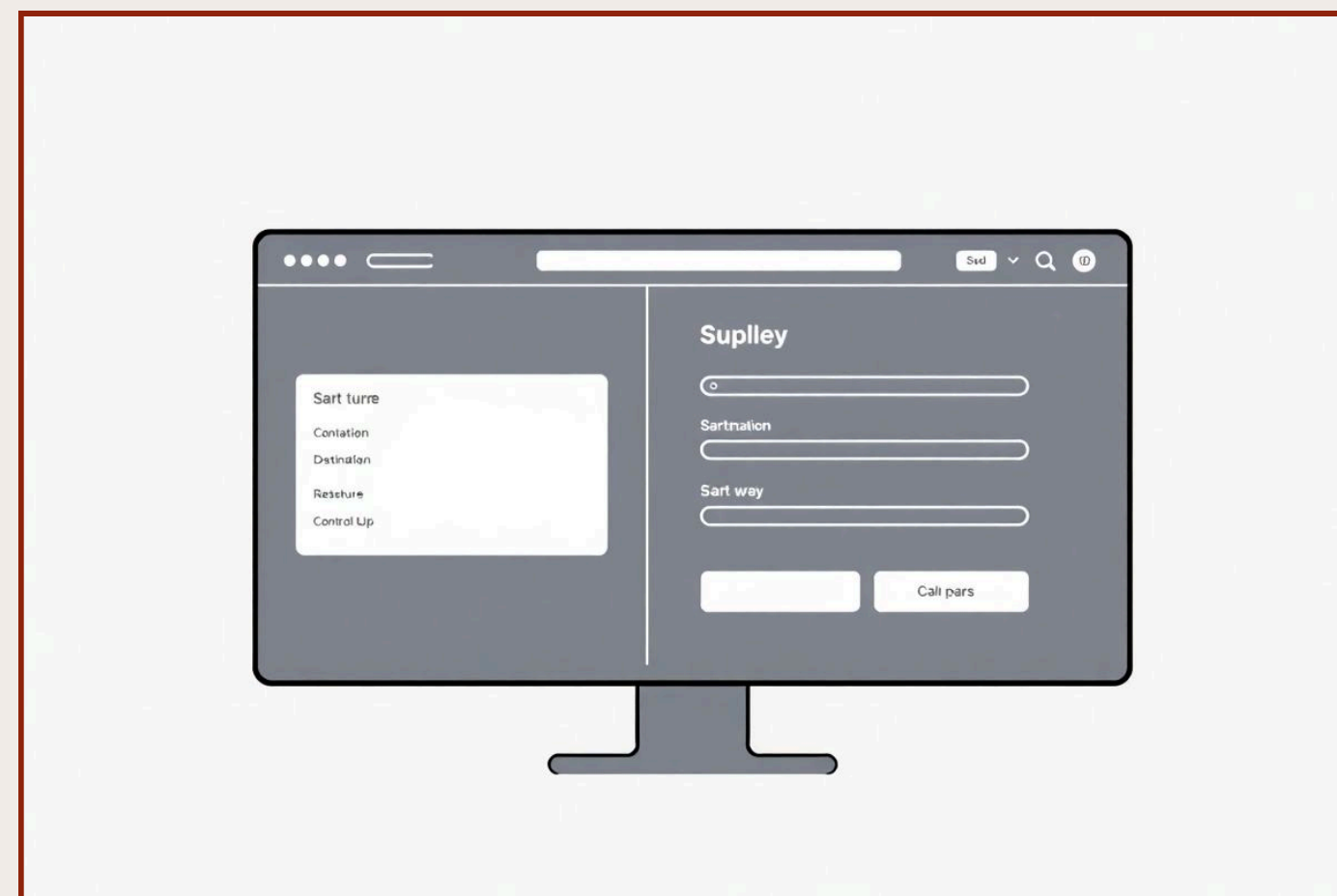
- **Frontend UI:** Responsive Next.js interface for route planning and map visualization using TailwindCSS.
- **Mapping Module:** Visualizes route polylines and safety markers using Leaflet.js and OpenStreetMap.
- **Routing Module:** Integrates OSRM API to calculate path geometries, travel distance, and duration.
- **Database Module:** Stores and retrieves real-time accident black spot data using Firebase Firestore.
- **Authentication Module:** Manages user roles and admin privileges for data moderation via Firebase Auth.
- **Generative AI Module:** Processes route hazards into conversational safety briefings using Genkit and Google Gemini.

SOFTWARE REQUIREMENT SPECIFICATION



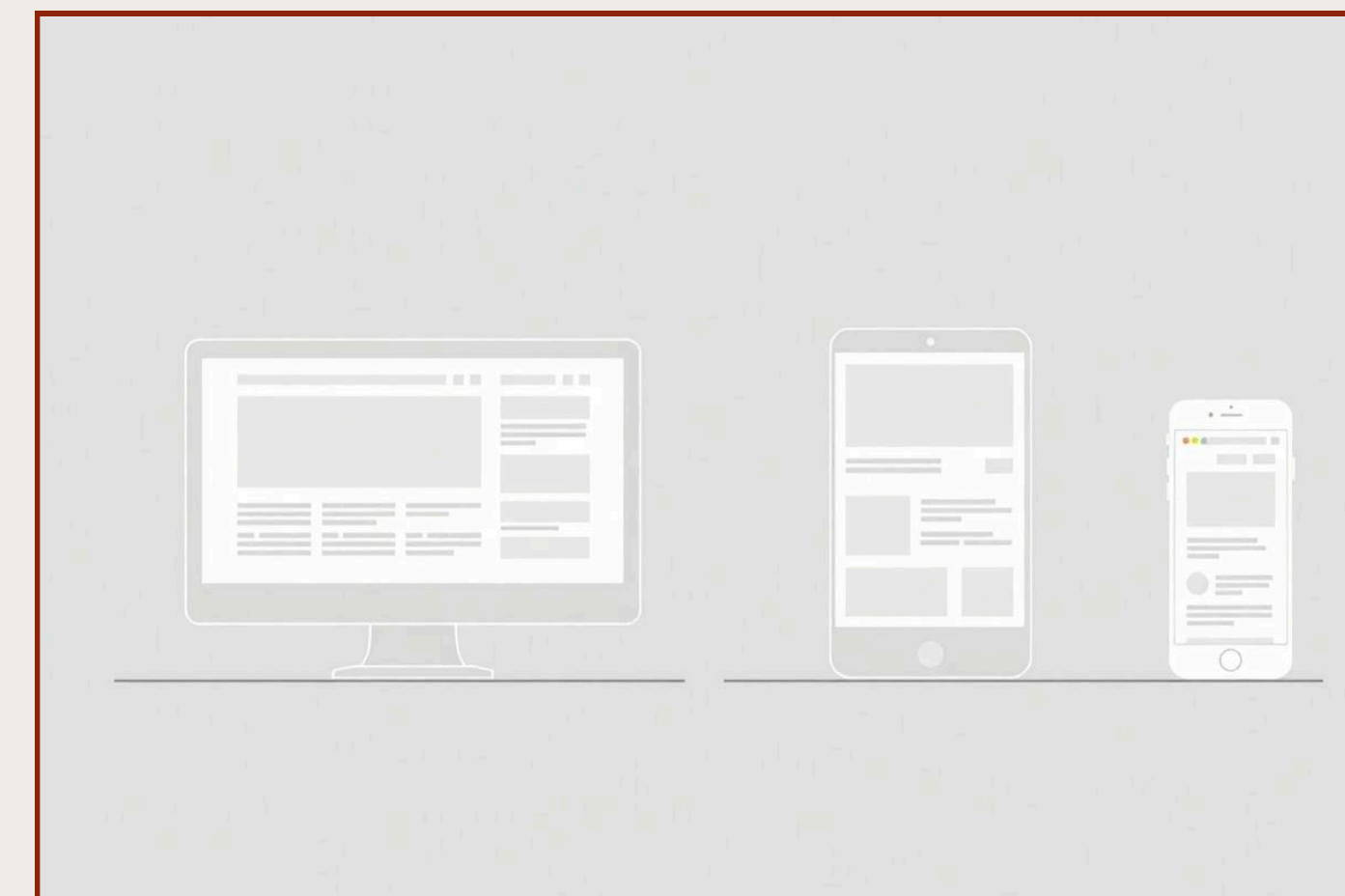
UI REQUIREMENTS

Responsive web interface for desktop and mobile.



SRS DOCUMENT

Covers functional and non-functional requirements, and system design.



CONTROL PANEL

Input for start, destination, and stops; interactive map display.

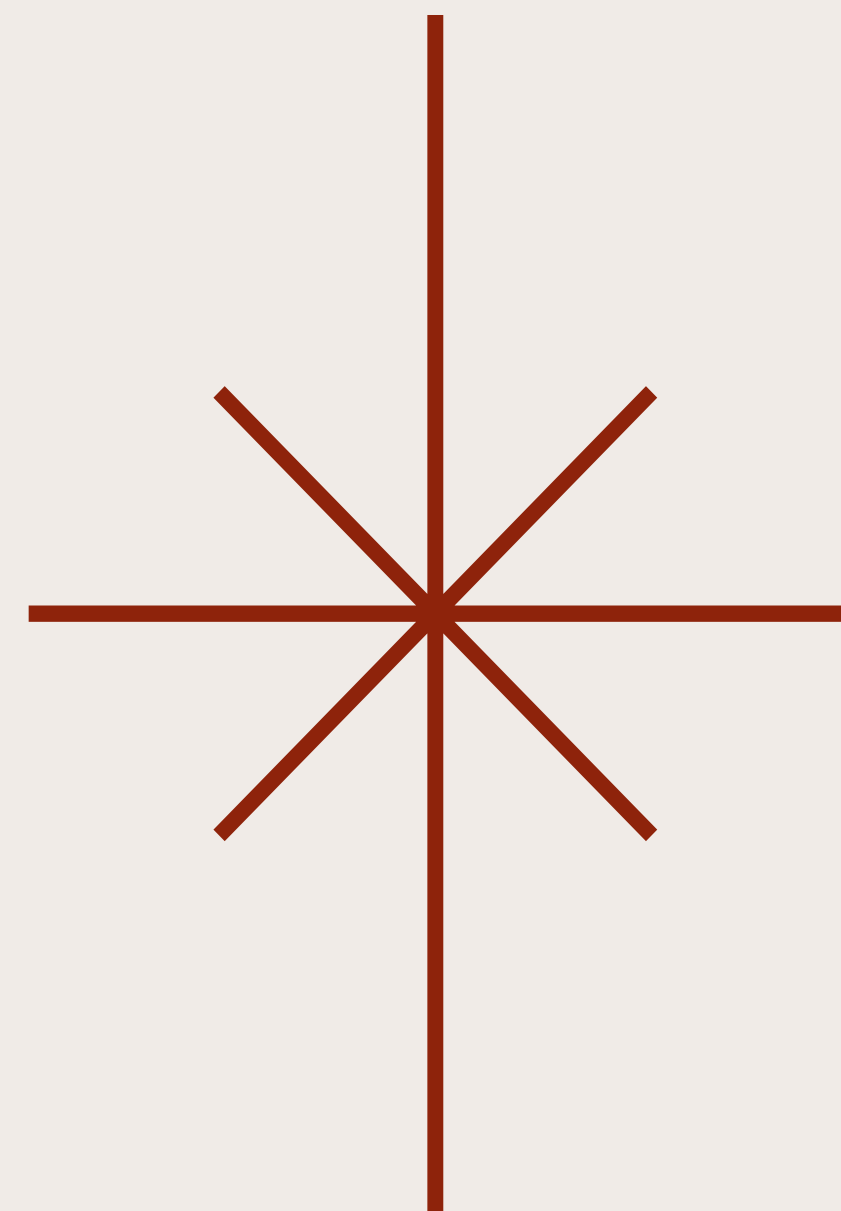
USER INTERFACE

- The application provides a responsive web-based user interface.
- The UI is developed using Next.js and styled with Tailwind CSS.
- Option to add accident prone areas.
- Basic validation and user feedback.



SOFTWARE REQUIREMENTS

- Operating System: Windows / Linux / macOS
- Frontend Framework: Next.js
- Programming Language: TypeScript
- Styling Framework: Tailwind CSS
- Database & Backend Services: Firebase
- Package Manager: npm
- Web Browser: Chrome, Edge, or Firefox



HARDWARE REQUIREMENTS

- A computer, tablet, or smartphone capable of running a modern web browser.
- For real-time navigation, the device must have GPS/geo location capabilities.



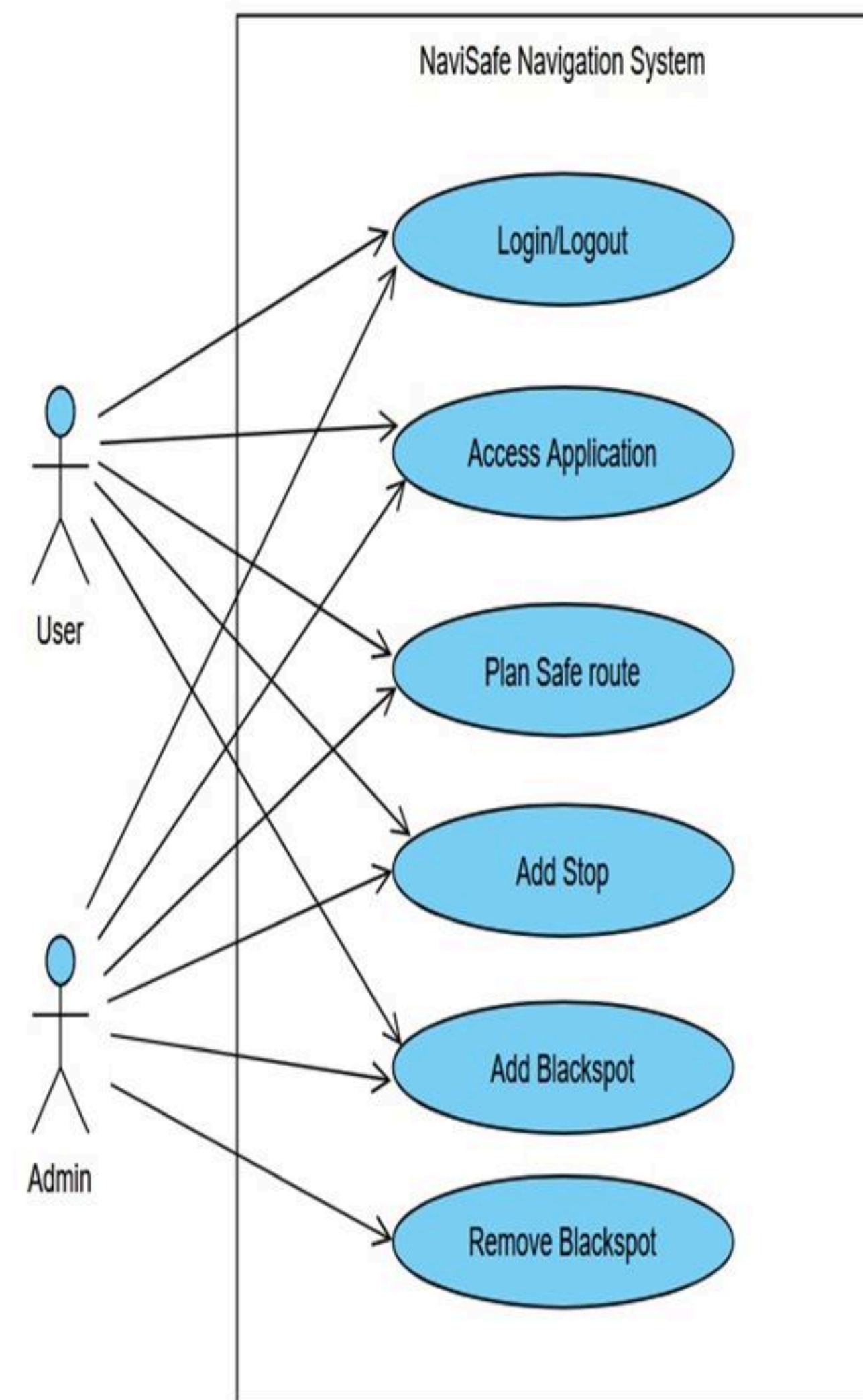
FUNCTIONAL REQUIREMENTS

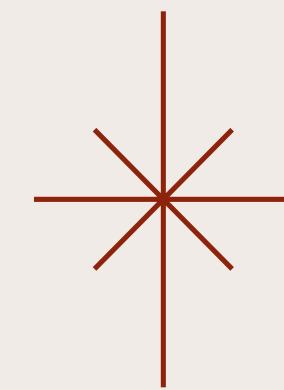
- **Route Planning:** Input start and destination locations to calculate travel paths.
- **Map Visualization:** Display active routes and path geometries dynamically on the map.
- **Hazard Markers:** Visualize accident-prone "black spots" using interactive map icons.
- **AI Safety Briefing:** Generate natural language warnings for hazards found along the route.
- **Turn-by-Turn Navigation:** Enable a live navigation mode for real-time driver guidance.
- **Live Trip Metrics:** Show current speed, remaining distance, and ETA during transit.
- **Proximity Alerts:** Trigger toast notifications when within 100 meters of a danger zone.

FUNCTIONAL REQUIREMENTS

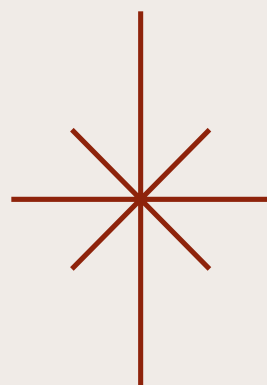
- **Crowd sourced Reporting:** Allow users to report new black spots by clicking the map.
- **Report Confirmation:** Enable users to verify and increment the counter for existing hazards.
- **Data Moderation:** Provide administrators with the ability to delete incorrect black spot data.
- **Travel Modes:** Support routing specifically optimized for either Cars or Bikes.
- **Multi-Stop Routing:** Allow the addition of intermediate waypoints to a single journey.

USE CASE DIAGRAM

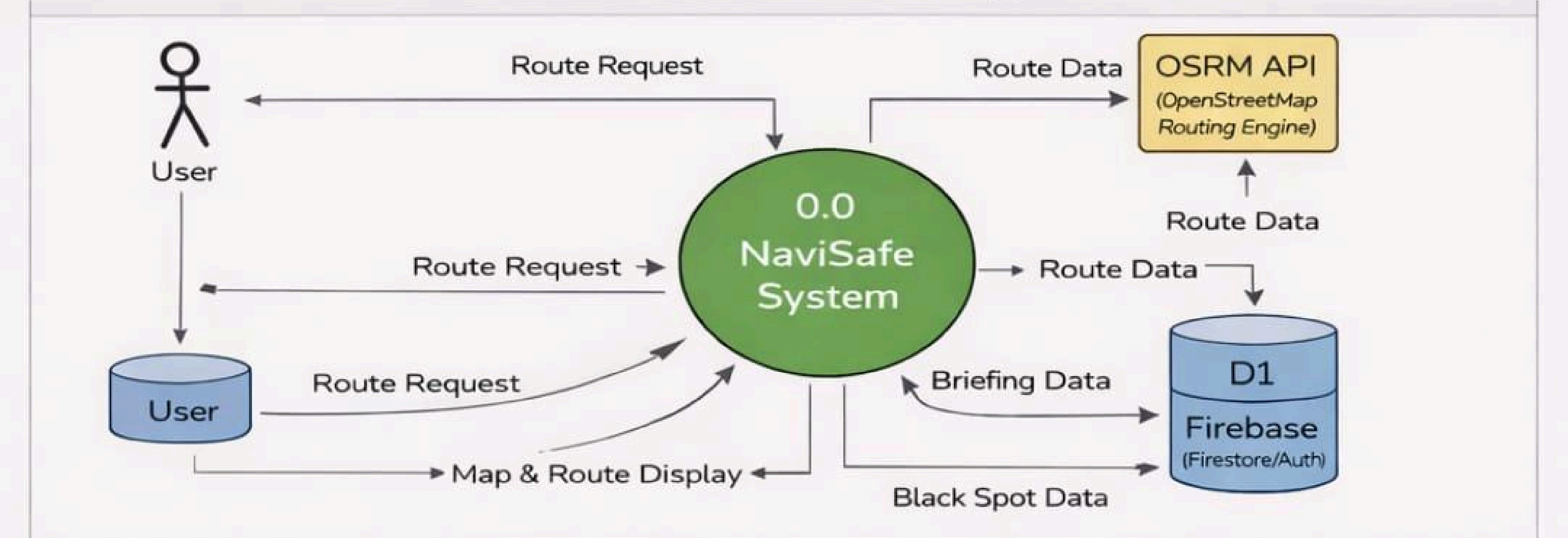




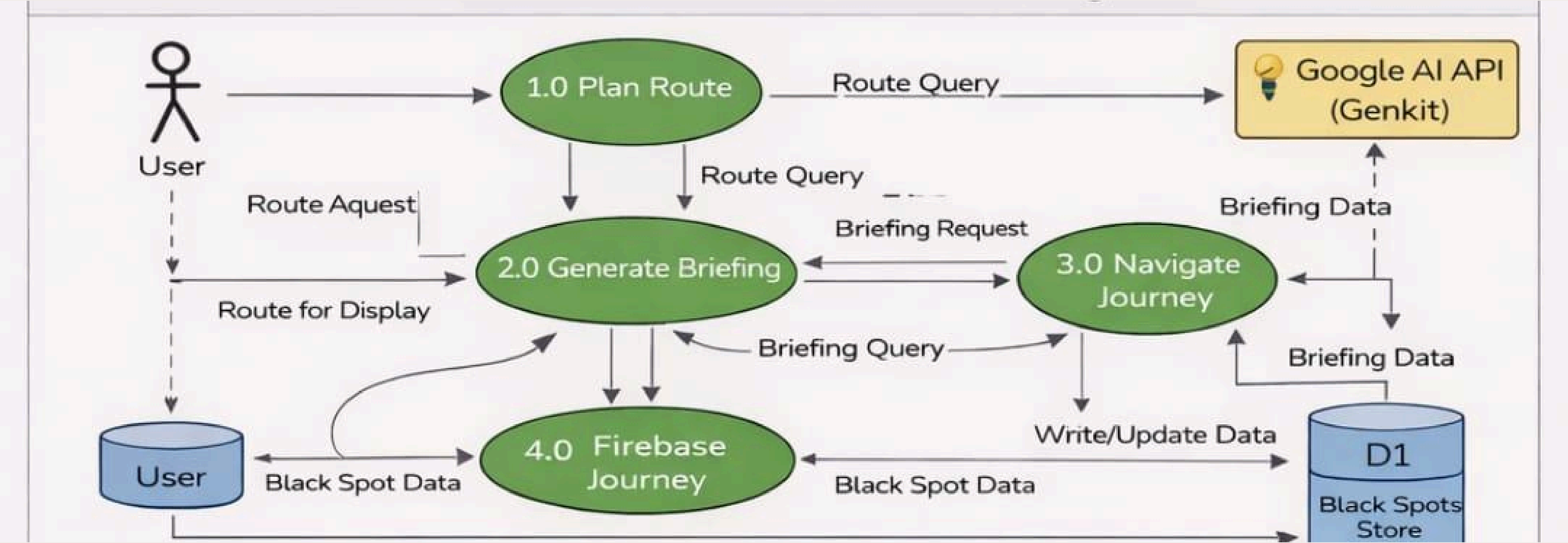
DATA FLOW DIAGRAM



LEVEL 0



LEVEL 1

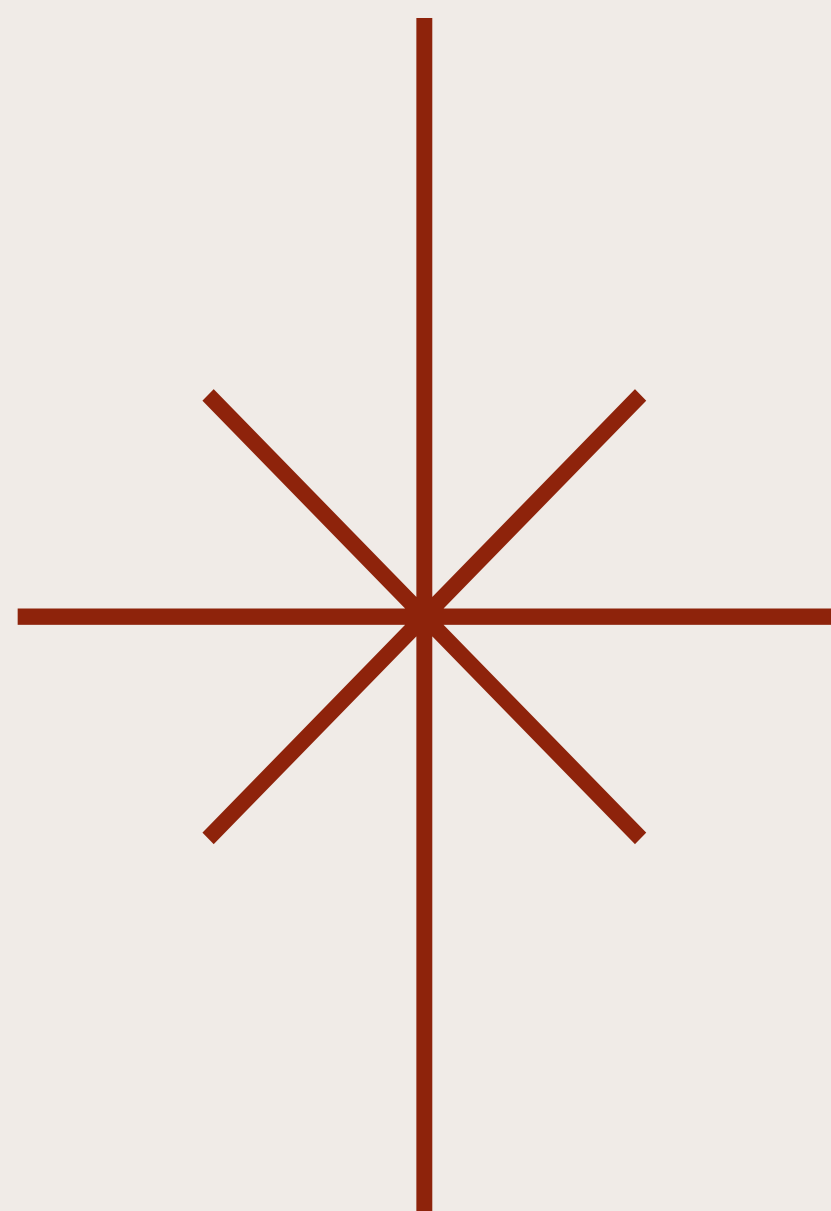


DATABASE REQUIREMENTS

- Database Type: Firebase Firestore (NoSQL)
- The database shall store application-related data.
- Data access shall be controlled using security rules.
- The database shall support real-time data operations.

NON-FUNCTIONAL REQUIREMENTS

- Performance: The system should load pages quickly.
- Security: Data access should be secure using Firebase rules.
- Scalability: The system should support future growth.
- Usability: The application should be easy to use.
- Reliability: The system should provide consistent availability.
- Responsiveness: The UI must adapt cleanly to a range of screen sizes, from mobile phones to desktop monitors.



RESULTS

User

Admin



User Login

Login as a standard user to plan routes and contribute data.

[Login as User](#)

User

Admin



Admin Login

Username

admin

Password

.....

Login

NaviSafe

AI-Powered Safety Navigation

Plan a Route

Car

Bike

Start Location

Destination

Plan Route

Contribute Data

Add on Map

Add by Coords

Please login to contribute data.

Powered By Group 2 S6C

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 **NaviSafe**
AI-Powered Safety Navigation



Plan a Route

 Car

 Bike

 Chengannur, Chengannur Railway



 Pandalam, Adoor, Pathanamthitta, Kerala,

 Plan Route

Contribute Data

 Add on Map

 Add by Coords

Please login to contribute data.

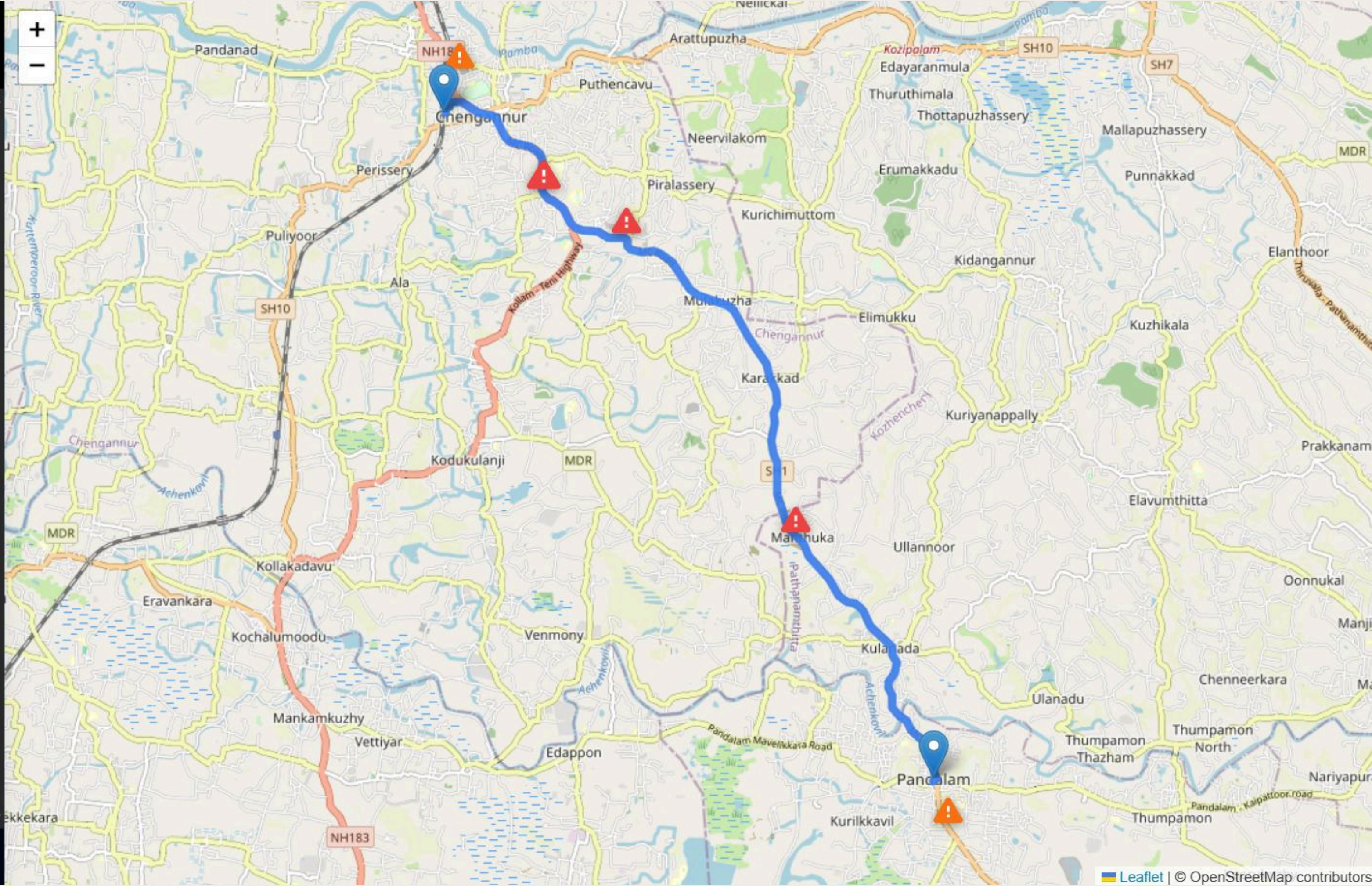
Route Ready



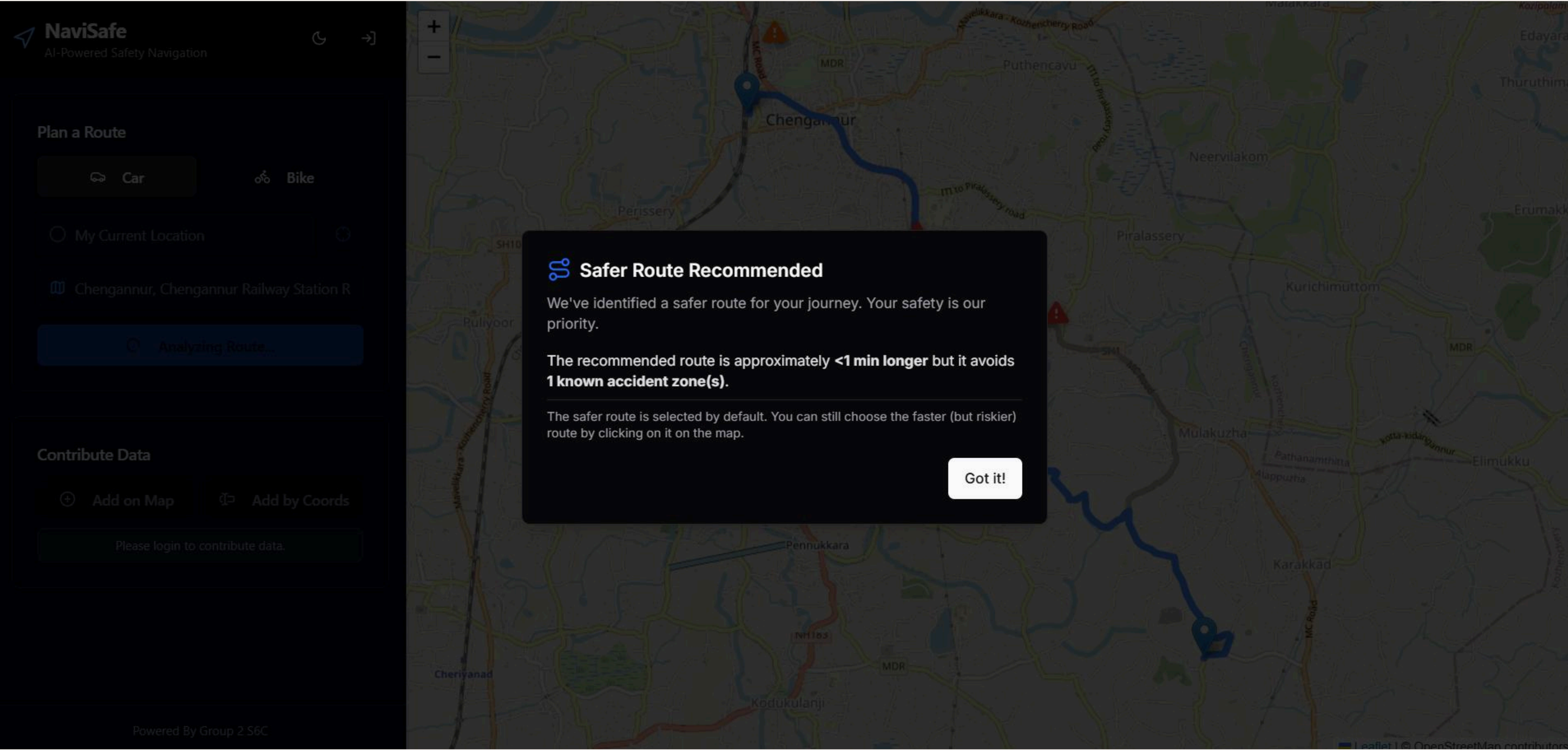


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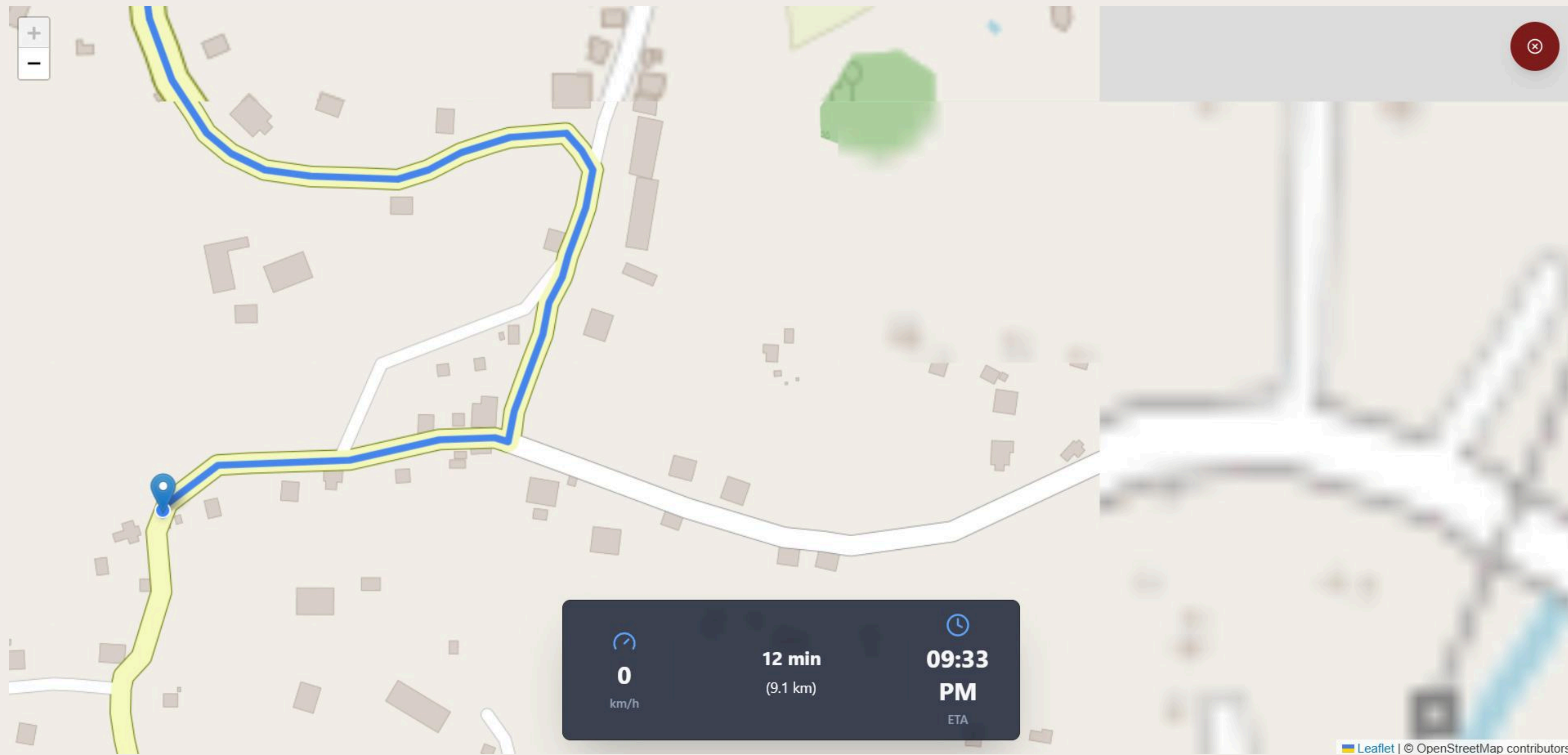


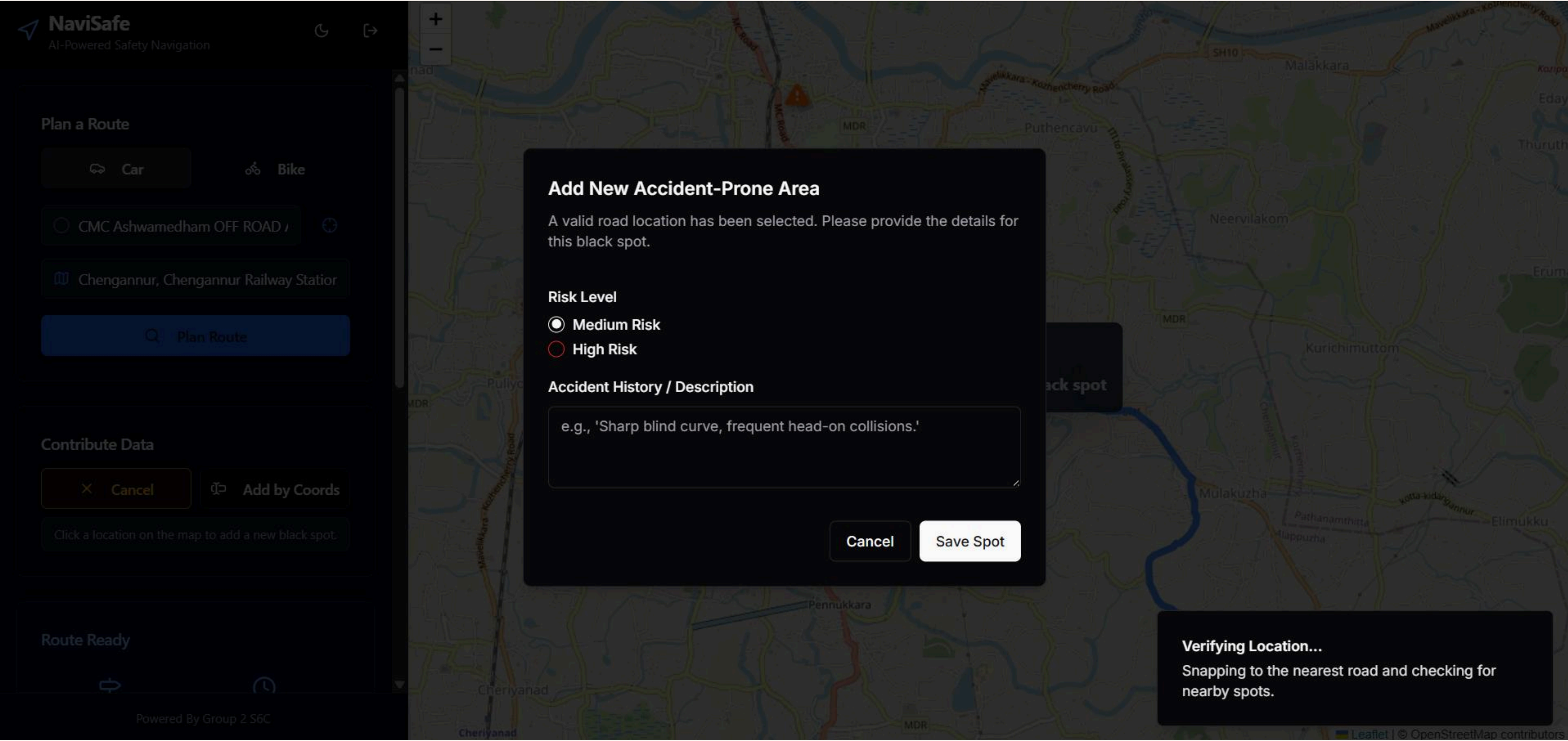


 Leaflet | © OpenStreetMap contributors



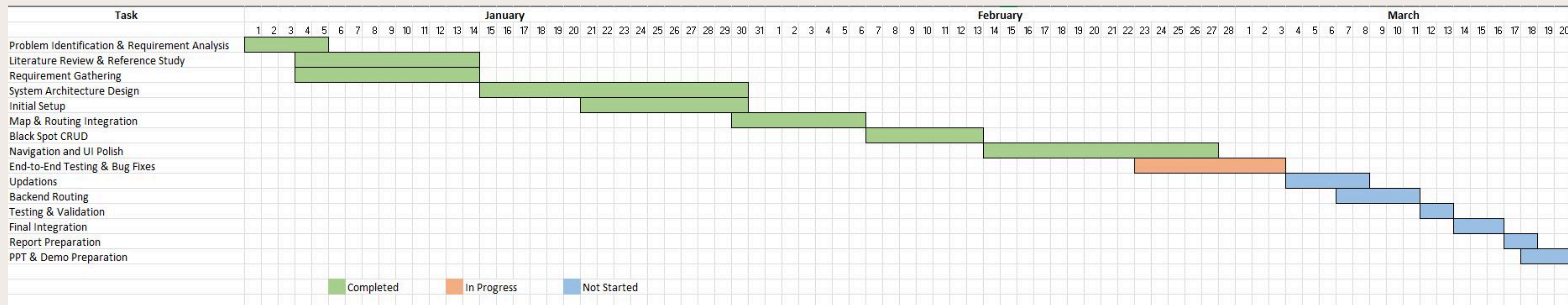
The screenshot displays the NaviSafe web application interface. On the left, a dark sidebar contains the NaviSafe logo and tagline 'AI-Powered Safety Navigation'. Below this, there are sections for 'Contribute Data' (with 'Add on Map' and 'Add by Coords' buttons and a login prompt), 'Route Ready' (showing a distance of 18.3 km and an estimated time of 22 minutes), and 'Stops' (listing 'Chengannur, Chengannur Railway...' as a stop 8.3 km from the start, with a search bar and 'Start'/'Clear' buttons). The main area on the right shows a map of the region around Kozhikode, India. A blue route is plotted from Muthoor in the north to Karakkad in the south. Along this route, several orange triangular warning icons with exclamation marks are placed, indicating traffic alerts or hazards. Key locations labeled on the map include Changanassery, Kumbanad, Kozhicherry, and Pandam. The bottom of the sidebar indicates the application is 'Powered By Group 2 S6C'.





The image shows a screenshot of the NaviSafe mobile application. The interface is dark-themed. On the left, there's a sidebar with the NaviSafe logo and tagline 'AI-Powered Safety Navigation'. Below this, there's a 'Plan a Route' section with buttons for 'Car' and 'Bike', input fields for 'Start Location' and 'Destination', and a 'Plan Route' button. Further down is a 'Contribute Data' section with 'Add on Map' and 'Add by Coords' buttons. The main area displays a map of Chengannur, India, with a red line indicating a route. A white pop-up box with a red warning icon and text 'High Risk Zone' is overlaid on the map, stating 'Accident prone area' and 'Reported by 2 user(s)', with a 'Remove Spot' button. The map shows various roads, including NH183 and MC Road, and landmarks like Chengannur Municipal Stadium. The bottom of the screen shows 'Powered By Group 2 S6C' and map data attribution for Leaflet and OpenStreetMap.

GANTT CHART



CONCLUSION

- NaviSafe addresses a critical gap in navigation safety.
- Combines accident analytics with real-time routing.
- Enhances safety for vulnerable road users.
- Supports the vision of smart and safe transportation systems.

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**THANK
YOU!**

